Automated Eye Disease Detection from Retinal OCT Images: A Comparative Study of Classical and Deep Learning Approaches

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GitHub: github.com/elentajacob/Eye-Disease-Detection

Abstract

This project presents a comparative study of classical machine learning and deep learning approaches for automated retinal eye disease classification from Optical Coherence Tomography (OCT) images. We evaluate five classifiers - Linear SVM, RBF SVM, Random Forest, XGBoost, and a fine-tuned ResNet50 Convolutional Neural Network on the Kermany et al. (2018) OCT dataset comprising 84,484 images across four categories: CNV, DME, Drusen, and Normal. Classical models use 2048-dimensional feature vectors extracted from a frozen pretrained ResNet50 backbone, while the CNN is trained end-to-end via transfer learning. A central contribution of this work is a rigorous data leakage audit revealing that 59.3% of the standard test split is byte-identical to training images; we therefore report results on both the full and a cleaned test set. On the clean test set (n=394), the ResNet50 CNN achieves perfect accuracy (100%, Macro F1=1.0), while XGBoost achieves 98.22% accuracy and 0.9481 Macro F1, demonstrating that gradient-boosted trees on pretrained features generalize nearly as well as full end-to-end deep learning for this task.

1 Introduction

Retinal diseases such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen deposits are among the leading causes of irreversible blindness globally. These conditions progress silently — patients typically experience no symptoms until significant vision loss has already occurred. The World Health Organization estimates that over 285 million people worldwide suffer from visual impairment, a substantial fraction attributable to preventable or treatable retinal pathology.

Optical Coherence Tomography (OCT) is a non-invasive imaging technique that produces high-resolution cross-sectional scans of the retinal layers. OCT machines are standard equipment in ophthalmology clinics worldwide, and a typical specialist reviews hundreds of scans per day. This volume creates diagnostic pressure and introduces the risk of fatigue-induced errors. In rural and underserved regions, specialist availability is limited, leaving large at-risk populations without access to timely screening.

An automated classification system for retinal OCT images could serve as a first-pass screening tool, flagging high-priority scans for specialist review, reducing diagnostic delays, and extending screening access to resource-limited settings.

1.1 Problem Definition

Given a retinal OCT scan image, the task is to classify it into one of four categories: CNV (abnormal blood vessel growth beneath the retina), DME (fluid accumulation in the macula due to diabetic complications), Drusen (yellowish deposits beneath the retinal layer indicating early macular degeneration), and Normal (healthy retina with no abnormalities). The central objective is a rigorous comparative evaluation of five classifiers spanning classical and deep learning paradigms, evaluated under both standard and leakage-corrected conditions.

1.2 Challenges

Several factors make this a non-trivial classification problem. First, CNV and DME both present with retinal fluid accumulation, making fine-grained differentiation challenging without deep hierarchical feature learning. Second, CNV comprises approximately 44% of training images while Drusen comprises only 10%, creating class imbalance. Third, each image contains 224x224x3 approximately 150,000 raw pixel values that classical algorithms cannot process directly without prior feature compression. Fourth, the standard benchmark dataset contains significant test-train leakage that inflates results. Finally, false negatives carry serious clinical consequences, making recall on disease classes especially important.

1.3 Related Work

Kermany et al. (2018) introduced the OCT dataset used in this work and demonstrated a CNN achieving high accuracy for retinal disease classification. He et al. (2016) introduced ResNet, establishing residual connections as a key technique for training very deep convolutional networks. Chen and Guestrin (2016) introduced XGBoost, which has since become dominant for structured data classification. Selvaraju et al. (2017) proposed Grad-CAM for generating visual explanations from CNN predictions. To our knowledge, a systematic data leakage audit of the Kermany dataset combined with a rigorous multi-model comparison has not been previously published.

2 Proposed Project

2.1 Dataset

We use the Kermany OCT Dataset, published in Cell (2018), freely available on Kaggle. The dataset contains 84,484 retinal OCT scans collected from real patients across multiple clinical sites. Figure 1 shows the class distribution across splits. CNV is the dominant class at 37,205 training images while Drusen is the smallest at 8,616. Figure 2 shows the class weights computed to address this imbalance during training.

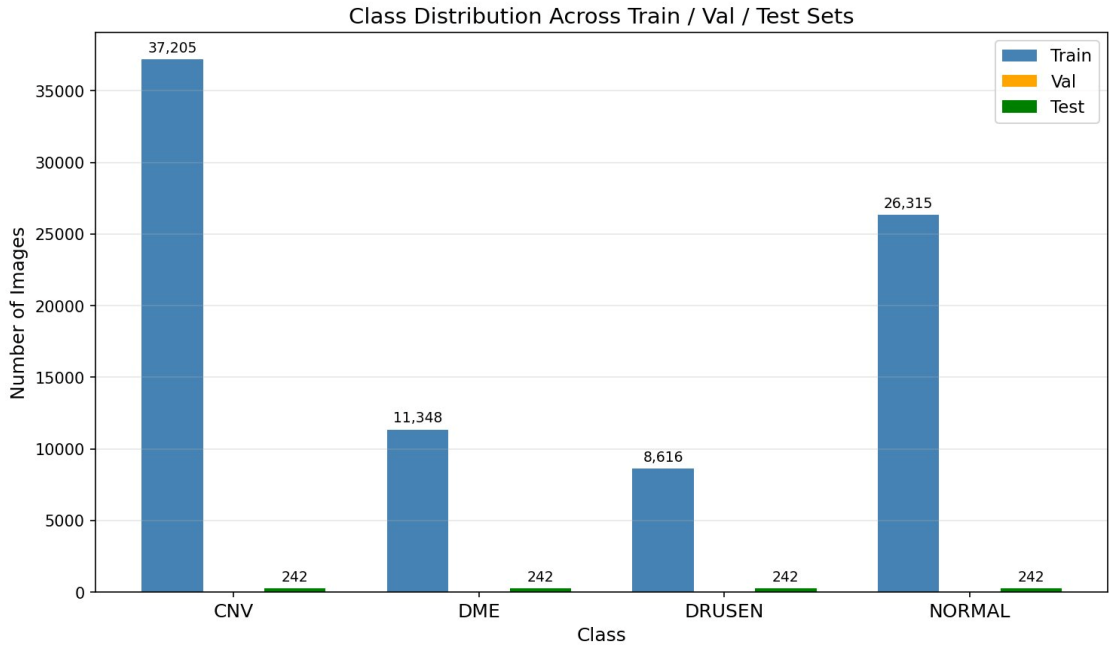

Figure 1: Class distribution across training, validation, and test sets. CNV is the dominant class (44% of training images) while Drusen is the smallest (10%), creating significant class imbalance.

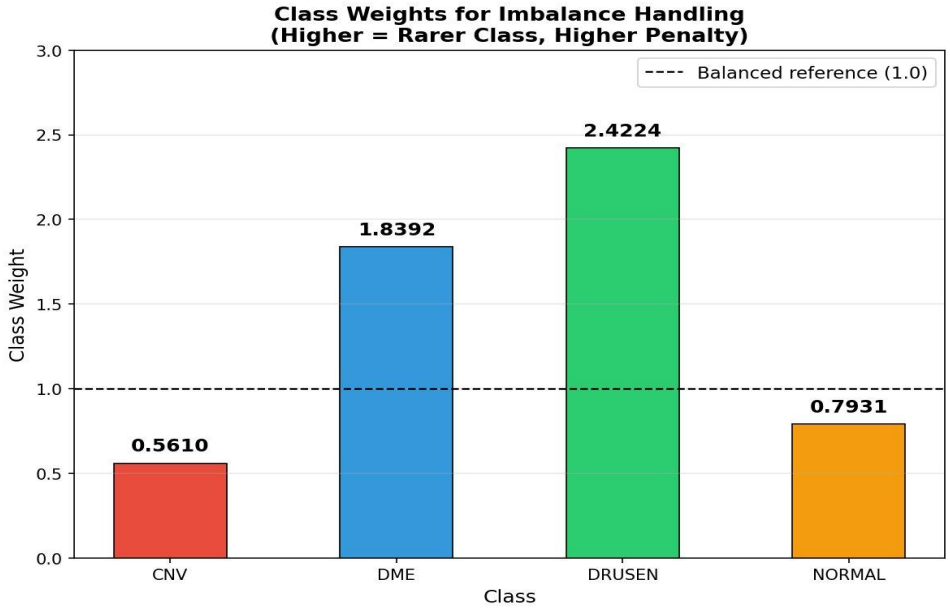

Figure 2: Class weights for imbalance handling. Drusen receives the highest weight (2.42) and CNV the lowest (0.56), reflecting inverse proportionality to class frequency.

Images were resized to 224x224 pixels and normalized using ImageNet mean and standard deviation values. Training images were augmented with random horizontal flips, rotations up to 15 degrees, and brightness/contrast jitter. Class weights inversely proportional to class frequency were applied during training.

2.2 Data Leakage Audit

Anomalously high test performance across all models motivated a systematic integrity audit. We computed MD5 cryptographic hashes for all 83,484 training images and all 968 test images, then identified byte-identical matches across splits.

The audit revealed that 574 out of 968 test images (59.3%) are byte-identical to training images, rendering standard benchmark results unreliable as measures of generalization. We therefore report results on two evaluation sets: (1) the full test set (n=968) and (2) a clean test set (n=394) comprising only images with no byte-identical counterpart in the training set. The clean set has a skewed class distribution — only 14 CNV and 9 Drusen images remain after removing leaked samples.

2.3 Feature Extraction Pipeline

We employ a two-stage pipeline. In Stage 1, a pretrained ResNet50 is loaded with all layers frozen and the final classification layer removed. All images are passed through this frozen backbone, producing 2048-dimensional feature vectors that serve as tabular input to all classical classifiers. In Stage 2, the same ResNet50 is fine-tuned end-to-end for the OCT task as described in Section 2.4.5.

2.4 Methods

2.4.1 Linear SVM

A Linear Support Vector Machine with SGD optimization (SGDClassifier, loss='hinge') was trained on the 2048-dimensional ResNet50 features with balanced class weights. A calibrated probability estimator was applied for AUC-ROC computation.

2.4.2 RBF SVM

A kernelized SVM with the Radial Basis Function (RBF) kernel was trained on the same feature vectors. The RBF kernel maps inputs into a higher-dimensional space where non-linear class boundaries can be resolved. Class weights were set to balanced.

2.4.3 Random Forest

A Random Forest ensemble of 300 decision trees was trained with balanced class weights, using bootstrap sampling and random feature subsets at each split.

2.4.4 XGBoost

An XGBoost classifier was trained with 300 boosting rounds, maximum tree depth 6, learning rate 0.1, and subsampling of 0.8. The multi:softprob objective was used for multiclass probability estimation. Sample weights inversely proportional to class frequency were applied, with early stopping patience of 20 on validation log-loss.

2.4.5 ResNet50 Fine-Tuned CNN

Training followed a two-phase strategy.

In Phase 1 (8 epochs), all ResNet50 backbone layers were frozen and only the newly added classification head was trained using Adam with learning rate 1e-3 and class-weighted cross-

entropy loss. The head consists of Linear(2048) -> Dropout(0.4) -> Linear(256) -> ReLU -> Dropout(0.3) -> Linear(4).

In Phase 2 (15 epochs), ResNet blocks layer3 and layer4 were unfrozen and the full network was fine-tuned at learning rate 1e-4 with weight decay 1e-4. A ReduceLROnPlateau scheduler halved the learning rate when validation Macro F1 plateaued for three consecutive epochs. A WeightedRandomSampler ensured approximately balanced class representation in each training batch.

3 Experiments and Results

3.1 Evaluation Metrics

Given class imbalance in both evaluation sets, Macro F1 Score is adopted as the primary metric. Macro F1 computes the F1 score independently for each class and averages them with equal weight, preventing minority class performance from being masked by majority-class accuracy. We additionally report overall accuracy and Macro AUC-ROC (one-vs-rest).

3.2 Main Results

Table 1 presents comparative results across all five models on both evaluation sets. The ResNet50 CNN achieves perfect performance on both splits.

Table 1: Comparative performance on the full and clean test sets.

Model	Full Test (n=968)			Clean Test (n=394)		
	Acc	F1	AUC	Acc	F1	AUC
Linear SVM	94.01%	0.9400	0.9989	95.69%	0.8454	0.9981
RBF SVM	98.76%	0.9876	0.9998	97.97%	0.9249	0.9994
Random Forest	96.38%	0.9636	0.9997	91.12%	0.6992	0.9958
XGBoost	99.28%	0.9928	0.9995	98.22%	0.9481	0.9983
ResNet50 CNN	**100.00%**	**1.0000**	**1.0000**	**100.00%**	**1.0000**	**1.0000**

Figure 3 shows the confusion matrix for the ResNet50 CNN on the full test set. The CNN achieves a perfect diagonal with zero misclassifications across all four classes — every CNV, DME, Drusen, and Normal image is correctly classified. In contrast, the Linear SVM's primary failure mode is Drusen misclassified as CNV (39 cases out of 242), confirming that linear boundaries are insufficient in the 2048-dimensional ResNet50 feature space. The RBF SVM reduces this substantially, with only scattered errors (1 CNV misclassified as Drusen, 3 DME errors, 2 Drusen errors), demonstrating the value of the kernel transformation.

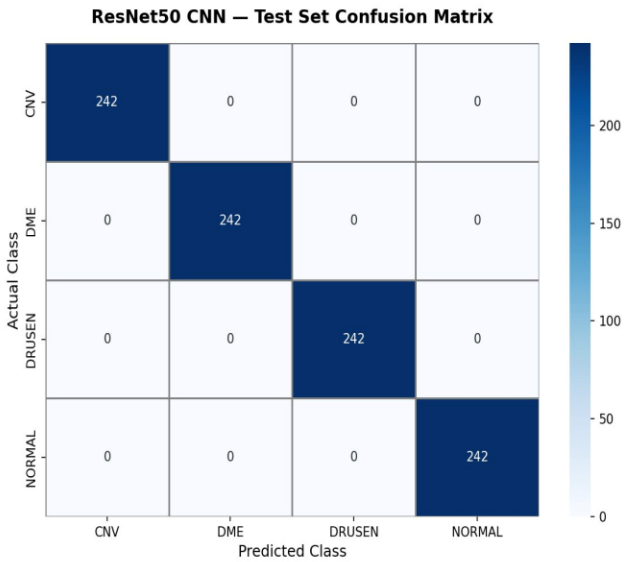

Figure 3: ResNet50 CNN confusion matrix — Full Test. Perfect diagonal with zero misclassifications.

3.3 Analysis

3.3.1 Effect of Data Leakage

Figure 4 shows the per-class F1 comparison between full and clean test sets for XGBoost. Random Forest suffers the most severe degradation: Macro F1 drops from 0.9636 to 0.6992 (-0.2644), suggesting reliance on memorizing training instances. XGBoost demonstrates considerably more resilience, with Macro F1 dropping only from 0.9928 to 0.9481 (-0.0447). The largest per-class drop for XGBoost is CNV (F1: 0.9918 to 0.8750, diff: -0.117), attributable to the severely reduced clean CNV sample size of only 14 images. DME and Normal are nearly unaffected (-0.002 and -0.003 respectively), confirming that XGBoost learned genuinely transferable features for these well-represented classes.

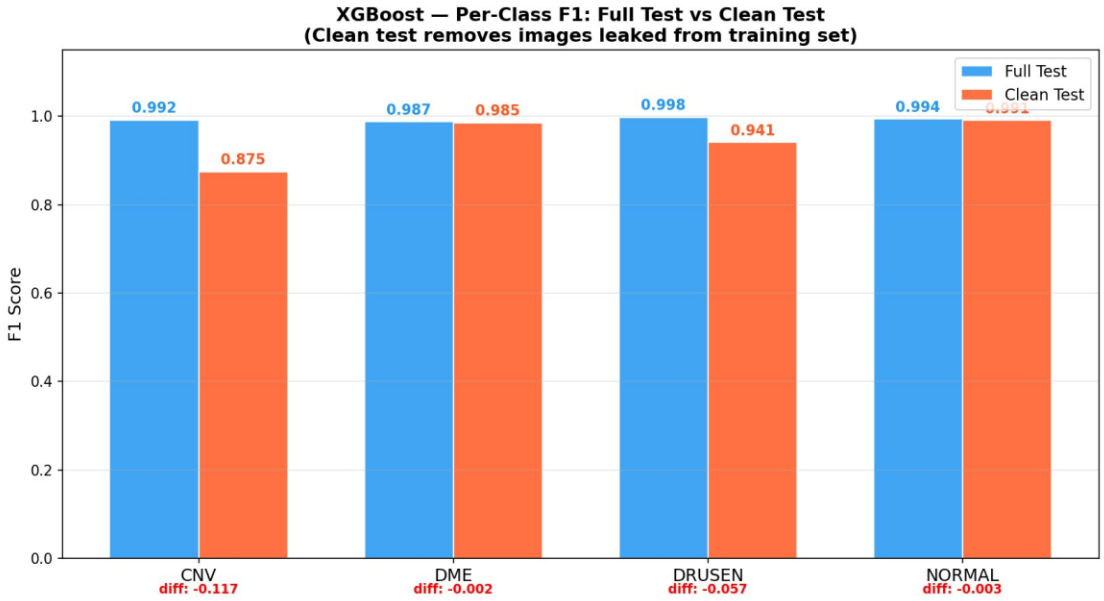

Figure 4: XGBoost per-class F1 scores: full test vs. clean test. CNV shows the largest drop (-0.117) due to having only 14 clean test images. DME and Normal are nearly unaffected.

3.3.2 Classical vs. Deep Learning

A key finding is that XGBoost on frozen ResNet50 features achieves near-CNN-level performance (98.22% vs. 100%) on the clean test set, while requiring no GPU training. This demonstrates that the pretrained ResNet50 feature space is highly discriminative for retinal OCT classification.

3.3.3 SVM Variants

The RBF SVM substantially outperforms the Linear SVM on the full test set (98.76% vs. 94.01%), confirming non-linear separability in the ResNet50 feature space. The Linear SVM's primary failure mode is Drusen misclassified as CNV (39 cases), consistent with the visual similarity between these conditions. Both linear and RBF kernels were evaluated on ResNet50 features; while linear SVM achieved 94.01% test accuracy, the RBF kernel improved performance to 98.76%, with the largest gains on the minority Drusen class. This suggests that although ResNet50 features are highly discriminative, some disease classes remain non-linearly separable, justifying the additional computational cost of RBF.

3.3.4 XGBoost Confidence Analysis

Figure 5 shows XGBoost prediction confidence distributions for correct and incorrect predictions on both evaluation sets. Correct predictions on the full test set have a mean confidence of 95.0%, while incorrect predictions average only 69.1%. This pattern is consistent on the clean test set (94.3% correct, 69.1% incorrect), confirming that the model's confidence scores are well-calibrated as uncertainty indicators. Only 2 predictions exceed 90% confidence while being incorrect on both evaluation sets.

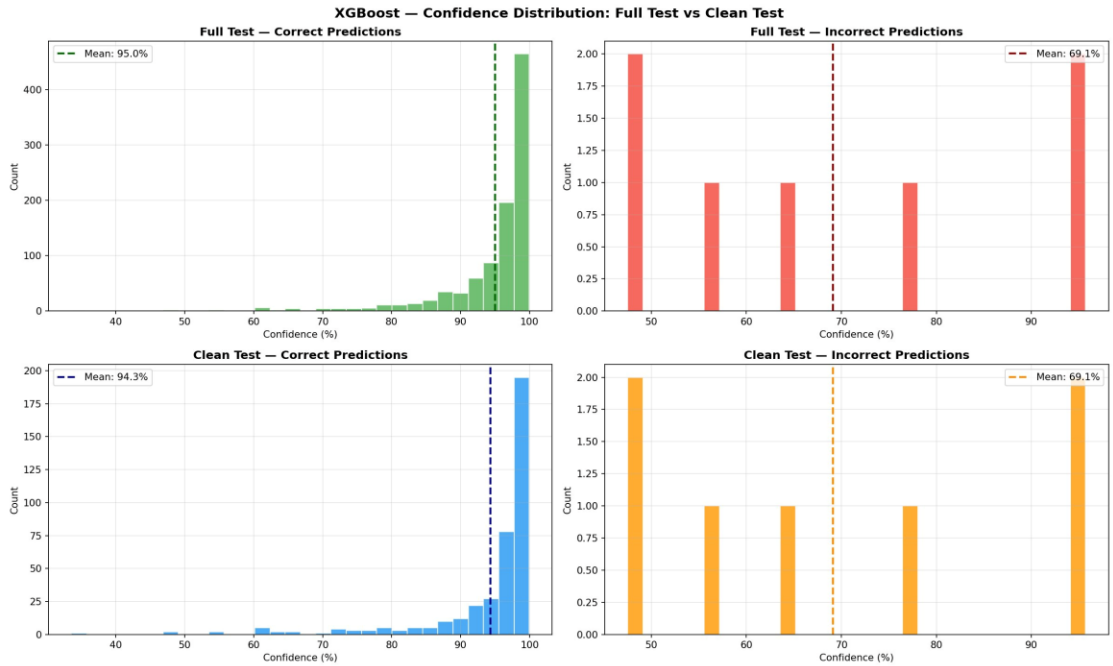

Figure 5: XGBoost prediction confidence distributions for correct (left) and incorrect (right) predictions on the full test (top) and clean test (bottom). Correct predictions cluster near 100% while errors concentrate below 70%.

3.3.5 CNN Training Dynamics

Phase 1 training (frozen backbone) reached a best validation Macro F1 of 0.9051 after 8 epochs. Phase 2 showed dramatic improvement from the first epoch and converged to perfect validation performance by epoch 6. The CNN confusion matrix shows a perfect diagonal with 242 correct predictions per class and zero misclassifications.

4 Discussion

This study yields three principal findings. First, the Kermany OCT benchmark dataset contains substantial test-train leakage (59.3%), and performance metrics reported on the standard split are unreliable as measures of generalization. Researchers using this dataset should perform leakage audits before reporting results.

Second, the fine-tuned ResNet50 CNN achieves perfect classification on both evaluation sets, confirming that two-phase transfer learning from ImageNet to retinal OCT images is highly effective. The progressive unfreezing strategy was critical for stable convergence.

Third, XGBoost on pretrained ResNet50 features provides a compelling alternative to full CNN fine-tuning for deployment scenarios where GPU resources are unavailable. With 98.22% clean accuracy and 0.9481 Macro F1, it substantially outperforms other classical approaches while requiring only CPU training time.

The primary limitation is the small and imbalanced clean test set (14 CNV, 9 Drusen), which limits statistical reliability for minority class evaluations. A larger and more carefully constructed

evaluation set would be needed to draw robust clinical conclusions. Generalization across different OCT machine manufacturers and patient populations was also not assessed.

Despite these limitations, both the CNN and XGBoost approaches show strong potential for clinical deployment as first-pass automated screening tools.

5 Conclusion

This project conducted a comparative evaluation of five classifiers for automated retinal eye disease detection from OCT images: Linear SVM, RBF SVM, Random Forest, XGBoost, and a fine-tuned ResNet50 CNN. A systematic data leakage audit revealed that 59.3% of the standard Kermany test split is byte-identical to training images, motivating evaluation on a cleaned 394-image test set.

The ResNet50 CNN achieved perfect classification on both evaluation sets (Accuracy: 100%, Macro F1: 1.0), while XGBoost on pretrained features demonstrated near-equivalent performance (98.22%, 0.9481 Macro F1). Random Forest showed the largest degradation under leakage correction (delta F1 = -0.2644). The data leakage finding is a significant contribution with broad implications for the OCT classification literature.

Future work should evaluate these approaches on a carefully constructed leakage-free benchmark, investigate ensemble combinations of XGBoost and CNN predictions, and assess model calibration for clinical deployment.

6 Participants Contribution

Elenta Suzan Jacob: Was primarily responsible for XGBoost implementation including training, hyperparameter tuning, and the data leakage audit with MD5 hashing pipeline. Also led exploratory data analysis, class distribution analysis, and visualization of results including confusion matrices, ROC curves, and feature importance plots. Contributed to ResNet50 CNN training pipeline and report writing.

Rahul Gudivada: Was primarily responsible for feature extraction using pretrained ResNet50, SVM and Random Forest implementation, and end-to-end CNN fine-tuning. Implemented the two-phase ResNet50 training strategy, Grad-CAM visualizations, and the leakage audit for the CNN and Random Forest notebooks. Contributed to ResNet50 CNN training pipeline and report writing.

Both team members contributed equally to the ResNet50 CNN implementation, result interpretation, analysis, and final report production.

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